Waleed Sherwani

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EDUCATION

Memorial University of Newfoundland (MUN) **2022 – Present**
Bachelors of Computer Engineering *St. John's, NL, Canada*

- Academic Term 5 (3rd Year) | Class of 2026 | GPA: 3.83

WORK EXPERIENCE

Data Analytics and Project Control Intern **Sep. 2023 – Present**
SNC-Lavalin - Dragados - Pennecon General Partnership (SDP-GP) *Argentia, NL, Canada*

- Employing Python for specialized data analytics workflows, using libraries such as Pandas, NumPy and Matplotlib for data manipulation and visualization, enhancing the efficiency of data processing and analysis.
- Leveraging the C++ language to create custom data analysis solutions, optimizing workflows with the integration of specialized libraries like Boost and Eigen for efficient data handling and mathematical operations.
- Developing and maintaining interactive Power BI dashboards, providing real-time project status updates to stakeholders, enhancing transparency and decision support.
- Preparing comprehensive project documentation, ensuring compliance with industry standards, protocols, and version control using Git.
- Applying advanced data analytics tools and techniques, including ML and AI algorithms, to identify trends, anomalies, and optimization opportunities within project workflows.

Data Quality Assurance Analyst **May. 2023 – Present**
Data and Image Analysis Group - Memorial University of Newfoundland *St. John's, NL, Canada*

- Analyzing medical images, ensuring data accuracy, and collaborating with researchers from both the DIAG and FAISAL Labs at the Memorial University of Newfoundland and the Simon Fraser University to improve patient care through automated image-derived metrics.
- Contributing to groundbreaking research in medical imaging and computational anatomy, gaining hands-on experience in a specialized field while supporting the development of machine-learning algorithms.

Robotics Research Intern **Jan. 2023 – May. 2023**
Memorial University of Newfoundland *St. John's, NL, Canada*

- Integrated DekaBot & Adafruit Color Sensors into ZumoBots and developed custom C++ software to operate them.
- Improved ZumoBots' implementation in Swarm Robotics by comparing evaluated improvements from the DekaBot.
- Wrote contextual Line Detection Software and Alignment Algorithms with PID Control loop mechanisms.
- Implemented I^2C protocol for communication between ZumoBots' microcontroller and various sensors and modules, improving the functionality and efficiency of the robots.
- Designed and Wired Specialized Proto boards for practical testing.
- Created PCB and Electrical Wiring designs in KiCad and TinkerCad.
- Assembled circuits with sensors, LEDs, and inertial motion components.
- Tested and debugged sensors and actuators in design implementations.

ISWEP Research Assistant **Jun. 2022 – Sep. 2022**
Graduate Office – Memorial University of Newfoundland *St. John's, NL, Canada*

- Programmed a Python script that parses data from an Excel workbook and performs a selective Internet Search.
- Designed a survey analysis format using Excel, with automatic, predefined data visualizations such as linear models, predicted changes, and bell curves.
- Created flowcharts and analyzed immigration programs for MUN international graduates.

EXTRA-CURRICULAR ACTIVITIES

Director of Programming – CELC 2024 Committee **Feb. 2023 – Present**

- Researching and identifying potential speakers to deliver engaging presentations on Engineering Leadership.
- Scheduling speakers' presentations, ensuring that their topics align with the conference's overall theme.
- Reaching out to potential speakers to discuss the conference's theme and requirements, and to invite them to participate in the event.

BOTS - Bio-inspired Robotics Group **Feb. 2023 – Present**

- Collaborating with a multi-disciplinary team in the development of swarm robots, that apply bio-inspired principles to act locally and move the world into a more desirable state.

Student Tutor **Sep. 2022 – Present**

Engineering One Success Centre – Memorial University of Newfoundland *St. John's, NL, Canada*

- Tutored for Programming, Design, Statics, Physics, Mathematics and Chemistry courses.
- Developed a summary package for Engineering and Math courses, for future tutoring work and instructions.
- Explained concepts to different learners, adjusting as needed.

CERTIFICATIONS

- **Google Data Analytics** **Jan. 2023**
- **Onshape Fundamentals: CAD Learning Pathway Completion** **Jan. 2023**
- **Tech Stewardship Practice Program (TSPP)** **Dec. 2022**
- **Workplace Hazardous Materials Information System (WHMIS)** **Aug. 2022**

SKILLS

- C++ Python Java JavaScript R (Visualization) SQL (Database) C#
- PSpice MATLAB MS Office Onshape TinkerCAD KiCAD Power BI
- Effective problem-solver and strong communicator
- Proven leader, capable of working alone or in a collaborative environment.
- Excellent time management and organization abilities

COURSES

- C++ Python Java JavaScript R (Visualization) SQL (Database) C#
- PSpice MATLAB MS Office Onshape TinkerCAD KiCAD Power BI
- Effective problem-solver and strong communicator
- Proven leader, capable of working alone or in a collaborative environment.
- Excellent time management and organization abilities

INTERESTS

- Sketching
- Image Manipulation
- Writing
- Worldbuilding